

Semantic Runtime Protocol: Core Submission Version

Anonymous ACL submission

Abstract

Long-horizon LLM systems often preserve context through prompt accumulation, summarization, or retrieval, but these strategies do not explicitly model semantic state as a runtime object. We present **Semantic Runtime Protocol (SRP)**, a minimal runtime abstraction for studying bounded semantic drift under repeated compression-recovery cycles. The paper asks one narrow question: whether explicit semantic-state management can improve finite-horizon stability while remaining token-efficient. We evaluate SRP against lightweight baselines under a reproducible qualification-gated experiment pipeline. The resulting claim is intentionally modest: SRP is a bounded runtime abstraction for semantic stability, not a semantic operating system.

1 Introduction

The central limitation in long-horizon LLM interaction is the lack of a formal runtime abstraction for semantic state. Existing prompt, memory, and retrieval systems manage tokens or stored context, but they do not make semantic transformation, validation, recovery, and update first-class runtime operations.

We study SRP as a semantic-state runtime abstraction with four operators:

- compression
- recovery
- validation
- update

The first paper is intentionally narrow: it tests whether explicit semantic-state management improves finite-horizon stability under repeated transformation, relative to lightweight baselines.

Contributions

1. SRP as a minimal semantic-state runtime abstraction.
2. Bounded semantic drift as the primary stability lens.
3. A reproducible qualification-gated evaluation pipeline.

Failure Boundary

SRP is not guaranteed to work if the semantic vocabulary is corrupted, validation is unreliable, recovery becomes non-invertible in practice, or cumulative drift exceeds the tolerance budget. The paper treats this boundary as part of the claim.

2 Related Work

SRP is positioned against four groups of prior work:

- prompt compression and context reduction
- memory-augmented and retrieval-based systems
- agent frameworks and trajectory-based pipelines
- reproducibility-oriented evaluation infrastructure

The key difference is that SRP treats semantic state as an explicit runtime object rather than as token history or retrieved text alone.

3 Runtime State and Formalization

We model semantic state at step t as:

$$S_t = (M_t, V_t, P_t) \quad (1)$$

where M_t is structured semantic memory, V_t is vocabulary state, and P_t is policy state. SRP

defines four operators:

$$C : S_t \rightarrow Z_t \quad (2)$$

$$R : Z_t \rightarrow S'_t \quad (3)$$

$$\text{Val} : (S_t, S'_t) \rightarrow [0, 1] \quad (4)$$

$$U : (S'_t, F_t) \rightarrow S_{t+1} \quad (5)$$

The formalization remains lightweight. We define observable behavior $O(S)$ and use task-relevant distance $d(O(S), O(S'))$ as semantic error. Cumulative semantic drift over repeated cycles is the main stability quantity:

$$\Delta_k = d(O(S_0), O(S_k)) \quad (6)$$

The paper uses bounded semantic drift as a practical notion for finite-horizon evaluation.

4 Experimental Setup

The evaluation uses one fixed workflow and a small set of methods:

- raw prompt
- summarization
- retrieval-based memory
- SRP

The main tasks are:

- multi-turn instruction consistency
- iterative compression-recovery cycles

The main metrics are:

- semantic drift
- task success
- token cost

The evaluation is qualification-gated and replayable. The experiment package includes:

- EQ gate
- runtime equivalence traces
- formal batch runs
- paper-ready tables
- paper-ready figures

B M C Base Base D Base S Base T Base L SRP D SRP S SRP T SRP L ΔD ΔS ΔT ΔL

Table 1: Comparison between Semantic Runtime Protocol (SRP) and the strongest non-SRP baseline for each backend, model, and cycle setting. The baseline is selected by task success, then semantic drift, then token cost. Δ denotes SRP minus the selected baseline. Lower drift, token cost, and latency are better, while higher task success is better. Bold indicates the stronger value within each SRP-baseline pair.

5 Results

The main empirical claim is compact:

SRP achieves the lowest semantic drift across cycle depths and occupies a favorable low-drift, low-token regime under the qualified formal batch.

The results are summarized with three linked views:

- a drift-over-cycles figure
- a token-cost-versus-drift Pareto frontier
- a contract-stability figure

Together, they show that SRP is not merely token-light or one-step stable; it preserves semantic behavior more consistently across repeated transformation while remaining competitive on efficiency.

The reader should interpret the evidence in a restrained way: SRP improves long-horizon semantic stability without claiming universal superiority or broad system replacement.

6 Discussion

The results support a narrow reading of SRP. The paper is about finite-horizon semantic stability, not a full semantic operating system. The main discussion points are:

- why semantic state differs from memory
- why bounded drift is more realistic than exact preservation
- why the first implementation remains minimal
- what would be required to scale the design further

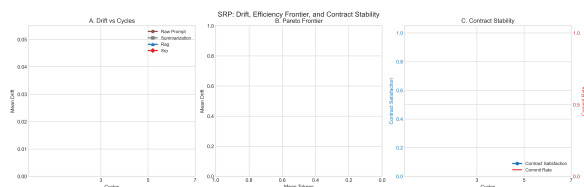


Figure 1: Three-panel main results. (A) Semantic drift across cycle depths for four methods. SRP consistently achieves the lowest drift and remains stable as cycle depth increases, while summarization degrades sharply and both raw prompt and RAG exhibit substantially higher drift. (B) Token cost versus drift Pareto frontier. SRP occupies a low-drift, low-token region, indicating that improvements are not attributable to increased computational budget. (C) SRP contract stability. Contract satisfaction remains stable across cycles, and commit decisions track semantic contract compliance rather than lexical similarity, suggesting that execution stability is governed by semantic consistency rather than surface-form preservation.

7 Limitations

The paper is intentionally narrow:

- finite-horizon evaluation
- proxy-based semantic metrics
- simplified state representation
- limited model and task scope

These constraints are part of the design, not accidental omissions.

8 Open Questions

The first paper preserves, rather than resolves, several questions:

- Is SRP primarily a runtime abstraction, a protocol, or a semantic communication layer?
- What is the minimal semantic-state object?
- Which drift metric should be considered the main one in the first paper?
- What failure cases should be shown explicitly?

9 Future Work

Future work can extend the present claim with:

- richer state schemas
- stronger validator design
- cross-model generalization

- semantic contract refinement

- benchmark expansion

- runtime lifecycle extensions such as freeze, merge, and delete

- transaction-style commit / rollback semantics

These are next-step extensions, not requirements for the first paper.

10 Submission Strategy

The most plausible near-term venues are workshop papers, short papers, student research tracks, or a technical report plus advisor feedback. The goal is a reproducible, bounded-scope paper with one strong figure and one careful framing, not a maximal full-scale systems paper.

11 Evidence Pack

This submission version is backed by the following qualified artifacts:

- EQ report
- runtime equivalence traces
- formal batch results
- drift plot
- contract/commit plot
- 3-panel main figure
- quality / efficiency / guardrail / camera-ready tables

If the paper is compressed further, the corresponding evidence should remain accessible through the project overview and paper-sections index rather than being deleted.

References

- Boyuan Chen, Mingzhi Wen, Yong Shi, Dayi Lin, Gopi Krishnan Rajbahadur, and Zhen Ming Jiang. 2022. [Towards training reproducible deep learning models](#). *arXiv preprint arXiv:2202.02326*.
- Shaolong Guo, Yuntao Wang, Ning Zhang, Zhou Su, Tom H. Luan, Zhiyi Tian, and Xuemin Shen. 2024. [A survey on semantic communication networks: Architecture, security, and privacy](#). *arXiv preprint arXiv:2405.01221*.

190	Huiqiang Jiang, Qianhui Wu, Chin-Yew Lin, Yuqing	247
191	Yang, and Lili Qiu. 2023. Llmlingua: Compressing	248
192	prompts for accelerated inference of large language	249
193	models . <i>arXiv preprint arXiv:2310.05736</i> .	250
194	Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio	251
195	Petroni, Vladimir Karpukhin, Naman Goyal, Hein-	252
196	rich Küttler, Mike Lewis, Wen-tau Yih, Tim Rock-	253
197	täschel, Sebastian Riedel, and Douwe Kiela. 2020.	254
198	Retrieval-augmented generation for knowledge-	255
199	intensive nlp tasks . In <i>Advances in Neural Inform-</i>	
200	<i>ation Processing Systems</i> .	
201	Barys Liskavets, Maxim Ushakov, Shuvendu Roy,	
202	Mark Klibanov, Ali Etemad, and Shane Luke. 2025.	
203	Prompt compression with context-aware sentence	
204	encoding for fast and improved llm inference . In	
205	<i>Proceedings of the AAAI Conference on Artificial</i>	
206	<i>Intelligence</i> .	
207	Sheng Lu, Hendrik Schuff, and Iryna Gurevych. 2023.	
208	How are prompts different in terms of sensitivity?	
209	<i>arXiv preprint arXiv:2311.07230</i> .	
210	Rui Meng, Song Gao, Dayu Fan, Haixiao Gao, Yining	
211	Wang, Xiaodong Xu, Bizhu Wang, Suyu Lv, Zhidi	
212	Zhang, Mengying Sun, Shujun Han, Chen Dong, Xi-	
213	aofeng Tao, and Ping Zhang. 2025. A survey of	
214	secure semantic communications . <i>arXiv preprint</i>	
215	<i>arXiv:2501.00842</i> .	
216	Loc X. Nguyen, Avi Deb Raha, Pyae Sone Aung, Dusit	
217	Niyato, Zhu Han, and Choong Seon Hong. 2025.	
218	A contemporary survey on semantic communica-	
219	tions: Theory of mind-based semantic communica-	
220	tion, generative ai-driven semantic communication,	
221	and deep joint source-channel coding . <i>arXiv preprint</i>	
222	<i>arXiv:2502.16468</i> .	
223	Zhuoshi Pan, Qianhui Wu, Huiqiang Jiang, Menglin	
224	Xia, Xufang Luo, Jue Zhang, Qingwei Lin, Victor	
225	Rühle, Yuqing Yang, Chin-Yew Lin, H. Vicky Zhao,	
226	Lili Qiu, and Dongmei Zhang. 2024. Llmlingua-	
227	2: Data distillation for efficient and faithful prompt	
228	compression . <i>arXiv preprint arXiv:2403.12968</i> .	
229	Amirhossein Razavi, Mina Soltangheis, Negar	
230	Arabzadeh, Sara Salamat, Morteza Zihayat, and	
231	Ebrahim Bagheri. 2025. Benchmarking prompt	
232	sensitivity in large language models . <i>arXiv preprint</i>	
233	<i>arXiv:2502.06065</i> .	
234	Harald Semmelrock, Tony Ross-Hellauer, Simone	
235	Kopeinik, Dieter Theiler, Armin Haberl, Stefan Thal-	
236	mann, and Dominik Kowald. 2024. Reproducibility	
237	in machine learning-based research . <i>arXiv preprint</i>	
238	<i>arXiv:2406.14325</i> .	
239	Lianlei Shan, Shixian Luo, Zezhou Zhu, Yu Yuan, and	
240	Yong Wu. 2025. Cognitive memory in large language	
241	models . <i>arXiv preprint arXiv:2504.02441</i> .	
242	Moritz Staudinger, Wojciech Kusa, Florina Piroi, Aldo	
243	Lipani, and Allan Hanbury. 2024. A reproducibility	
244	and generalizability study of large language model-	
245	based boolean query generation for systematic re-	
246	views . <i>arXiv preprint arXiv:2411.14914</i> .	
	Julian Junyan Wang and Victor Xiaoqi Wang. 2025. As-	247
	sessing consistency and reproducibility in the outputs	248
	of large language models in finance and accounting .	249
	<i>arXiv preprint arXiv:2503.16974</i> .	250
	Qingyue Wang, Yanhe Fu, Yanan Cao, Shuai Wang,	251
	Zhiliang Tian, and Liang Ding. 2023a. Recur-	252
	sively summarizing enables long-term dialogue mem-	253
	ory in large language models . <i>arXiv preprint</i>	254
	<i>arXiv:2308.15022</i> .	255
	Weizhi Wang, Li Dong, Hao Cheng, Xiaodong Liu,	256
	Xifeng Yan, Jianfeng Gao, and Furu Wei. 2023b.	257
	Augmenting language models with long-term mem-	258
	ory . <i>arXiv preprint arXiv:2306.07174</i> .	259
	Yaxiong Wu, Sheng Liang, Chen Zhang, Yichao Wang,	260
	Yongyue Zhang, Huifeng Guo, Ruiming Tang, and	261
	Yong Liu. 2025. A survey on memory mechanisms	262
	in the era of llms . <i>arXiv preprint arXiv:2504.15965</i> .	263